

11.31 Solution: Given E_ρ and B_z , $\mathbf{D}$ and $\mathbf{H}$ inside the cylinder can be expressed as

$$\mathbf{D}_{\text{in}} = \varepsilon \mathbf{E} + \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) [\beta^2 \mathbf{E}_\perp + \boldsymbol{\beta} \times \mathbf{B}] = \hat{\rho} \left[\gamma^2 \left(\varepsilon - \frac{\beta^2}{\mu} \right) E_\rho + \beta \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) B_z \right],$$

and

$$\mathbf{H}_{\text{in}} = \frac{\mathbf{B}}{\mu} + \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) [-\beta^2 \mathbf{B}_\perp + \boldsymbol{\beta} \times \mathbf{E}] = \hat{z} \left[-\gamma^2 \left(\beta^2 \varepsilon - \frac{1}{\mu} \right) B_z - \beta \gamma^2 \left(\varepsilon - \frac{1}{\mu} \right) E_\rho \right].$$

Here, the velocity is in the $\hat{\phi}$ direction, $\mathbf{E} = E_\rho \hat{\rho}$ and $\mathbf{B} = B_z \hat{z}$ are perpendicular to it. Outside of the cylinder, $\mathbf{D}_{\text{out}} = 0$ and $\mathbf{H}_{\text{out}} = B_0 \hat{z}$. From the continuity of the normal component of $\mathbf{D}$ and the tangent component of $\mathbf{H}$, we must have $D_\rho = 0$ and $H_z = B_0$. The first condition leads to

$$E_\rho = -\frac{\beta (\varepsilon - 1/\mu) B_z}{\varepsilon - \beta^2/\mu}.$$

Substitute this into the second condition, we will obtain

$$\left[-\left(\beta^2 \varepsilon - \frac{1}{\mu} \right) B_z + \beta (\varepsilon - 1/\mu) \frac{\beta (\varepsilon - 1/\mu) B_z}{\varepsilon - \beta^2/\mu} \right] \gamma^2 B_z = B_0,$$

which gives B_z as

$$B_z = \mu B_0 \frac{1 - \beta^2/\mu\varepsilon}{1 - \beta^2} = \mu B_0 \frac{1 - \omega^2 \rho^2/(c^2 \mu\varepsilon)}{1 - \omega^2 \rho^2/c^2}.$$

Then,

$$E_\rho = -\frac{\beta(1 - 1/(\mu\varepsilon))}{1 - \beta^2/(\mu\varepsilon)} B_z = -\frac{\beta(1 - 1/(\mu\varepsilon))}{1 - \beta^2} \mu B_0 = -\frac{\mu\omega\rho B_0}{c(1 - \omega^2 \rho^2/c^2)} \left(1 - \frac{1}{\mu\varepsilon} \right).$$

The voltage difference between the inner and outer surfaces is given by

$$\begin{aligned} V &= \left| \int_a^b E_\rho d\rho \right| = \frac{\mu\omega B_0}{c} \left(1 - \frac{1}{\mu\varepsilon} \right) \int_a^b \frac{\rho d\rho}{1 - \omega^2 \rho^2/c^2} \\ &= \frac{\mu\omega B_0}{c} \left(1 - \frac{1}{\mu\varepsilon} \right) \left(-\frac{c^2}{2\omega^2} \log(1 - \omega^2 \rho^2/c^2) \right) \Big|_a^b \\ &= \frac{\mu c B_0}{2\omega} \left(1 - \frac{1}{\mu\varepsilon} \right) \log \left(\frac{1 - \omega^2 a^2/c^2}{1 - \omega^2 b^2/c^2} \right). \end{aligned}$$

In the nonrelativistic limit, $\omega\rho \ll c$, we can approximate

$$\log(1 - \omega^2 \rho^2/c^2) \approx -\frac{\omega^2 \rho^2}{c^2}.$$

Then, in this limit, the voltage difference becomes

$$V = \frac{\mu\omega B_0}{2c} \left(1 - \frac{1}{\mu\varepsilon} \right) (b^2 - a^2).$$